
GENERAL RELATIVITY

Tutorial problem set 16, 22.01.2021, with solutions.

■ **PROBLEM 39** The ergosphere and ergoregion of the Kerr black hole.

- (a) Determine the boundary of the region in Kerr space-time inside of which it is impossible for an observer to remain at fixed coordinates (r, θ, ϕ) . Express this boundary in a form $r(\theta)$. Note that this region extends outside of the event horizon at $r_+ = MG_N + \sqrt{M^2 G_N^2 - a^2}$. The region of spacetime outside of the horizon where there are no stationary observers possible is called the ergoregion and its boundary is called the ergosphere. It is an effect of the rotation of the black hole. Sketch the ergoregion, ergosphere and the horizon.
- (b) Although static observers are not possible inside the ergosphere, observers with fixed just r and θ coordinates that rotate with constant angular velocity as seen from infinity ($d\phi/dt = \Omega = \text{const.}$) along with the black hole are allowed. Show that it is so.
- (c) Show that the minimal angular velocity of the massless particle at the horizon r_+ is

$$\Omega_H = \left(\frac{d\phi}{dt} \right)_{\min}(r_+) = \frac{a}{r_+^2 + a^2}. \quad (39.1)$$

- (d) Show that the Killing vector $\chi^\mu = K^\mu + \Omega_H R^\mu$ (where K^μ and R^μ are defined in problem (38)) has a Killing horizon, and determine where it is.

SOLUTION: (a) Let us recall the Kerr metric,

$$\begin{aligned} ds^2 = & - \left[1 - \frac{2G_N M r}{\rho^2} \right] dt^2 - \frac{4G_N M a r \sin^2(\theta)}{\rho^2} dt d\phi + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\theta^2 \\ & + \left[r^2 + a^2 + \frac{2G_N M r a^2 \sin^2(\theta)}{\rho^2} \right] \sin^2(\theta) d\phi^2. \end{aligned} \quad (39.2)$$

To determine the stationary limit surface we are interested in the two dimensional $t - \phi$ section of the metric (39.2),

$$\begin{aligned} d\sigma^2 &= g_{tt} dt^2 + 2g_{t\phi} dt d\phi + g_{\phi\phi} d\phi^2, \\ g_{tt} &= - \left(1 - \frac{r_S r}{\rho^2} \right), \quad g_{t\phi} = - \frac{r_S a r \sin^2(\theta)}{\rho^2}, \quad g_{\phi\phi} = \left[r^2 + a^2 + \frac{r_S r a^2 \sin^2(\theta)}{\rho^2} \right] \sin^2(\theta), \end{aligned} \quad (39.3)$$

where $r_S = 2G_N M$, $\rho^2 = r^2 + a^2 \cos^2(\theta)$, $\Delta = r^2 + a^2 - r_S r = \rho^2 + a^2 \sin^2(\theta) - r_S r$. Light moving on trajectories satisfying, $dr = 0 = d\theta$, obeys $d\sigma^2 = 0$ in (39.3), which can be also written as,

$$g_{tt} \dot{t}^2 + 2g_{t\phi} \dot{t} \dot{\phi} + g_{\phi\phi} \dot{\phi}^2 = 0 \implies \left(\frac{d\phi}{dt} \right)^2 + 2 \frac{g_{t\phi}}{g_{\phi\phi}} \left(\frac{d\phi}{dt} \right) + \frac{g_{tt}}{g_{\phi\phi}} = 0, \quad (39.4)$$

which is solved by,

$$\begin{aligned} \left(\frac{d\phi}{dt}\right)_{\pm} &= -\frac{g_{t\phi}}{g_{\phi\phi}} \pm \sqrt{\left(\frac{g_{t\phi}}{g_{\phi\phi}}\right)^2 - \frac{g_{tt}}{g_{\phi\phi}}} = \frac{1}{g_{\phi\phi}} \left[-g_{t\phi} \pm \sqrt{g_{t\phi}^2 - g_{\phi\phi}g_{tt}}\right] \\ &= \frac{r_S a r \sin(\theta) \pm \rho^2 \sqrt{\Delta}}{[(r^2 + a^2)\rho^2 + r_S r a^2 \sin^2(\theta)] \sin(\theta)}. \end{aligned} \quad (39.5)$$

where we made use of, $g_{t\phi}^2 - g_{\phi\phi}g_{tt} = \Delta \sin^2(\theta)$. At the stationary limit surface one of the angular velocity, $(\frac{d\phi}{dt})_{-}$, vanishes. From Eq. (39.5) we see that this happens when, $r_S a r \sin(\theta) - \rho^2 \sqrt{\Delta} = 0$. Finding roots of this equation is hard in general, since it leads to an equation of sixth order for $r(\theta)$. One can simplify the problem however, by noting that Eq. (39.5) can be also written as,

$$\left(\frac{d\phi}{dt}\right)_{\pm} = -\frac{g_{t\phi}}{g_{\phi\phi}} \left[1 \pm \sqrt{1 - \frac{g_{tt}g_{\phi\phi}}{g_{t\phi}^2}}\right], \quad (39.6)$$

from where we see that when,

$$\frac{g_{tt}g_{\phi\phi}}{g_{t\phi}^2} = -\frac{(\rho^2 - r_S r) [(r^2 + a^2)\rho^2 + r_S r a^2 \sin^2(\theta)]}{r_S^2 r^2 a^2 \sin^2(\theta)} = 0, \quad (39.7)$$

which is satisfied when,

$$r^2 - r_S r + a^2 \cos^2(\theta) = \left(r - \frac{r_S}{2}\right)^2 + a^2 \cos^2(\theta) - \frac{r_S^2}{4} = 0 \implies r_{\pm}^{\text{ES}} = \frac{r_S}{2} \pm \sqrt{\frac{r_S^2}{4} - a^2 \cos^2(\theta)}. \quad (39.8)$$

Inserting this into (39.5) one gets,

$$\left(\frac{d\phi}{dt}\right)_{-} = 0, \quad \left(\frac{d\phi}{dt}\right)_{+} = \frac{2a}{2a^2 \sin^2(\theta) + r_S r_{\pm}^{\text{ES}}}, \quad (39.9)$$

where we made use of, $\Delta(r_{\pm}^{\text{ES}}) = a \sin(\theta)$ and $\rho^2(r_{\pm}^{\text{ES}}) = r_S r_{\pm}^{\text{ES}}$.

The positive (negative) sign in the roots in (39.8) refers to the boundary of the external (internal) stationary limit surface, or ergosphere. The external ergoregion is placed in between the outer event horizon $r_+ = \frac{r_S}{2} + \sqrt{\frac{r_S^2}{4} - a^2}$ and the outer stationary limit surface at, $r_+(\theta) = \frac{r_S}{2} + \sqrt{\frac{r_S^2}{4} - a^2 \cos^2(\theta)}$. An alternative way to obtain the stationary limit surface is to consider the Killing horizon of the Killing vector field $K = \partial_t$, defined by

$$K^2 = g_{tt} = -\frac{r^2 - r_S r + a^2 \cos^2(\theta)}{\rho^2} = 0 \implies r_{\pm}^{\text{ES}} = \frac{r_S}{2} \pm \sqrt{\frac{r_S^2}{4} - a^2 \cos^2(\theta)}. \quad (39.10)$$

This avenue is explored in Carroll's textbook, and the result agrees with r_+^{ES} in Eq. (39.8). The outer ergosphere is a curve, and it is shown in figure 1.

(b) If θ and r are fixed, then from Eq. (39.9) we see that observers with

$$0 < \Omega(\theta) < \frac{2a}{2a^2 \sin^2(\theta) + r_S r_+^{\text{ES}}(\theta)}, \quad (r = r_+^{\text{ES}}) \quad (39.11)$$

are allowed at $r = r_+^{\text{ES}}$. Deeper inside the black hole, but still at $r > r_+$, from Eq. (39.5) we see that an observer can have angular velocity in the range,

$$\frac{r_S a r \sin(\theta) - \rho^2 \sqrt{\Delta}}{[(r^2 + a^2)\rho^2 + r_S r a^2 \sin^2(\theta)] \sin(\theta)} < \Omega(r, \theta) < \frac{r_S a r \sin(\theta) + \rho^2 \sqrt{\Delta}}{[(r^2 + a^2)\rho^2 + r_S r a^2 \sin^2(\theta)] \sin(\theta)}. \quad (39.12)$$

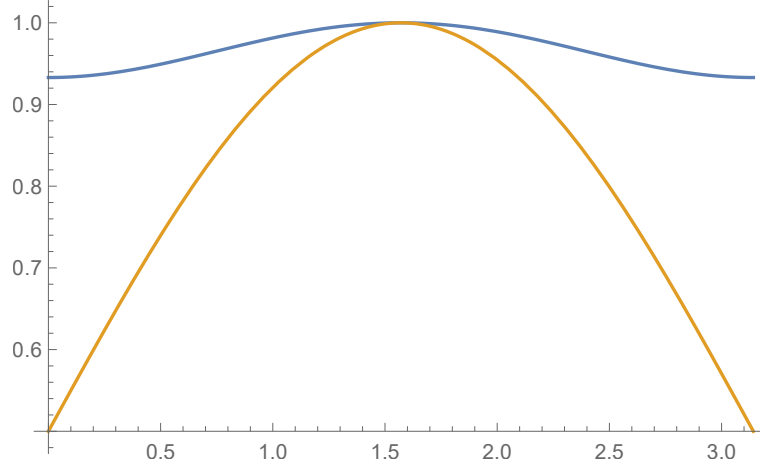


Figure 1: The ergosphere r_+^{ES}/r_S as a function of $\theta \in [0, \pi]$ for $2a/r_S = 1/2$ (blue) and $2a/r_S = 1$ (orange). The radius at $\theta = 0$ and $\theta = \pi$ is equal to the radius of the outer event horizon.

(c) Upon inserting $r_+^2 + a^2 = r_S r_+$ and $\Delta(r_+) = 0$ into Eq. (39.5) one obtains that, at $r = r_+$, the two angular velocities of light are equal to each other, and

$$\Omega_H = \left(\frac{d\phi}{dt} \right)_{\pm} = \frac{a}{r_+^2 + a^2} = \frac{a}{r_S r_+}, \quad (39.13)$$

which agrees with (39.1).

(d) The Killing horizon of $\chi^\mu = K^\mu + \Omega_H R^\mu = (1, 0, 0, \Omega_H)$ is given by,

$$\chi^2 = g_{tt}(\chi^t)^2 + 2g_{t\phi}\chi^t\chi^\phi + g_{\phi\phi}(\chi^\phi)^2 = g_{tt} + 2g_{t\phi}\Omega_H + g_{\phi\phi}\Omega_H^2 = 0, \quad (39.14)$$

Multiplying this by ρ^2 gives,

$$[1 + a^2\Omega_H^2 \sin^4(\theta)] [r^2 + a^2 - r_S r] - \sin^2(\theta) [a^2 - 2r_S r a \Omega_H + (r^2 + a^2)^2 \Omega_H^2] = 0. \quad (39.15)$$

The first two terms vanish at the event horizons of the Kerr black hole given by, $r = r_{\pm} = \frac{r_S}{2} \pm \sqrt{\frac{r_S^2}{4} - a^2}$. The term multiplying $\sin^2(\theta)$ vanishes at $r = r_+$, but it does not vanish at $r = r_-$ (it would vanish if one would change Ω_H to $a/(r_-^2 + a^2)$), showing that the Killing horizon of χ is the outer event horizon of the Kerr black hole. Note that (39.15) is a quartic equation in r , so it contains three more solutions for $r = r(\theta)$, which are roots of a cubic equation. Write down that equation and try to determine whether it contains physical solutions. *Hint:* Notice that the quartic equation (39.15) can be written in the form, $(r - r_+)[r^3 + pr^2 + qr + s] = 0$ and determine p , q and s from whether you can study the roots.

■ PROBLEM 40 Gauge invariant perturbations.

Consider a slightly perturbed Minkowski space,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad (40.1)$$

where $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ is the canonical Minkowski metric, and $h_{\mu\nu}$ is a small perturbation, $h_{\mu\nu} \ll 1$ ($\forall \mu, \nu$). The perturbation can be decomposed into parts that transform differently under *spatial rotations*, which is given in the usual way as

$$t \rightarrow \tilde{t} = t, \quad x^i \rightarrow \tilde{x}^i = R^i_j x^j, \quad \mathbf{R} \cdot \mathbf{R}^\top = \mathbf{1}, \quad \det(\mathbf{R}) = 1. \quad (40.2)$$

- (a) Show that under spatial rotations: (i) h_{00} transforms as a scalar, (ii) h_{0i} transforms as a vector, (iii) h_{ij} transforms as a tensor.

According to the Helmholtz's theorem on vector calculus, h_{0i} can be decomposed into two distinct parts, a transverse (divergence-free) part, and a longitudinal (curl-free) part,

$$h_{0i} = B_i + \partial_i S, \quad (40.3)$$

where $\partial^i B_i = 0$ and $\nabla \times (\nabla S) = 0$, and where we have used the result that the curl-free part can be written as a gradient of some scalar function S . In a similar way, a 2-tensor under spatial rotations can be decomposed as

$$h_{ij} = 2\delta_{ij}\psi + 2\partial_i\partial_j E + \partial_i F_j + \partial_j F_i + h_{ij}^{TT}. \quad (40.4)$$

Here ψ and E are scalar functions, F_j is a transverse (divergence-free) vector, and h_{ij}^T is a *transverse traceless tensor*,

$$\partial^i h_{ij}^{TT} = 0, \quad (h^{TT})^i_i = 0. \quad (40.5)$$

It is also customary to relabel the scalar under spatial rotations,

$$h_{00} = 2\phi. \quad (40.6)$$

- (b) Convince yourself that this decomposition preserves the number of degrees of freedom originally contained in $h_{\mu\nu}$.
- (c) Under the infinitesimal diffeomorphism transformation (which is an active coordinate transformation, as opposed to a passive one), the metric perturbation transforms as

$$h_{\mu\nu}(x) \rightarrow \tilde{h}_{\mu\nu}(x) = h_{\mu\nu}(x) - \partial_\mu \xi_\nu(x) - \partial_\nu \xi_\mu(x), \quad (40.7)$$

where $h_{\mu\nu}$ and ξ_μ are of the same order.

Utilizing the fact that ξ_μ can be decomposed with respect to spatial rotations into a scalar ξ_0 , and a vector $\xi_i = \xi_i^T + \partial_i \xi^S$, derive the following transformation rules for $h_{00}, B_i, S, \psi, E, F_i$ and h_{ij}^T under gauge (diffeomorphism) transformation (40.7)

$$\phi(x) \rightarrow \tilde{\phi}(x) = \phi(x) - \partial_0 \xi_0(x), \quad (40.8)$$

$$S(x) \rightarrow \tilde{S}(x) = S(x) - \xi_0(x) - \partial_0 \xi^S(x), \quad (40.9)$$

$$\psi(x) \rightarrow \tilde{\psi}(x) = \psi(x), \quad (40.10)$$

$$E(x) \rightarrow \tilde{E}(x) = E(x) - \xi^S(x), \quad (40.11)$$

$$B_i(x) \rightarrow \tilde{B}_i(x) = B_i(x) - \partial_0 \xi_i^T(x), \quad (40.12)$$

$$F_i(x) \rightarrow \tilde{F}_i(x) = F_i(x) - \xi_i^T(x), \quad (40.13)$$

$$h_{ij}^{TT}(x) \rightarrow \tilde{h}_{ij}^{TT}(x) = h_{ij}^{TT}(x). \quad (40.14)$$

- (d) Show that, apart from h_{ij}^{TT} and ψ which are gauge invariant, there are two other combinations of quantities in (40.8)-(40.13) that are gauge invariant. Show that all these make six gauge invariant degrees of freedom in total (this makes sense, since four can be eliminated by fixing a gauge).
- (e) You were asked in the last homework problem set to show that for metric (40.1) the Riemann tensor is

$$R_{\mu\nu\rho\sigma} = \frac{1}{2} \left(\partial_\rho \partial_\nu h_{\mu\sigma} + \partial_\sigma \partial_\mu h_{\nu\rho} - \partial_\rho \partial_\mu h_{\nu\sigma} - \partial_\sigma \partial_\nu h_{\mu\rho} \right). \quad (40.15)$$

Show that the Ricci tensor is

$$R_{\mu\nu} = \frac{1}{2} \left(\partial_\mu \partial^\alpha h_{\alpha\nu} + \partial_\nu \partial^\alpha h_{\alpha\mu} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h \right), \quad (40.16)$$

where indices are defined to be raised by the Minkowski metric $\partial^\alpha = \eta^{\alpha\beta} \partial_\beta$, d'Alembertian is an ordinary flat space one $\square = \partial^2 = \eta^{\alpha\beta} \partial_\alpha \partial_\beta$, and the trace is $h = \eta^{\alpha\beta} h_{\alpha\beta}$.

- (f) Using the fact that the Einstein equation in vacuum reduces to $R_{\mu\nu} = 0$, write out all the equations in terms of $\phi, S, \psi, E, B_i, F_i$ and h_{ij}^{TT} and show that only the gauge invariant combinations of these quantities appear in the equations of motion.
- (g) Now we show that the only dynamical quantity in these equations is h_{ij}^{TT} .

By taking appropriate linear combinations of the components of the Einstein equation show that

$$\nabla^2 \psi = 0. \quad (40.17)$$

If we assume that the perturbation vanishes at infinity, this equation has only one solution (remember electrostatics). From here the equations and solutions for the other non-dynamical gauge invariant perturbations follow. Show that the only equation that remains is

$$\square h_{ij}^{TT} = 0. \quad (40.18)$$

Therefore, h_{ij}^{TT} represents the two physical degrees of freedom of gravity in the absence of matter.

SOLUTION: (a) From a previous problem, we know that, under linear gauge transformations, $x^\mu \rightarrow \tilde{x}^\mu(x) = x^\mu + \xi^\mu(x)$, $h_{\mu\nu}$ transform as,

$$h_{\mu\nu}(x) \rightarrow \tilde{h}_{\mu\nu}(x) = h_{\mu\nu}(x) - \partial_\mu \xi_\nu(x) - \partial_\nu \xi_\mu(x). \quad (40.19)$$

Infinitesimal rotations $\mathbf{R}(\epsilon)$ in the xy plane can be written as,

$$\mathbf{R} = \begin{pmatrix} \cos(\epsilon) & -\sin(\epsilon) & 0 \\ \sin(\epsilon) & \cos(\epsilon) & 0 \\ 0 & 0 & 1 \end{pmatrix} \simeq \begin{pmatrix} 1 & -\epsilon & 0 \\ \epsilon & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (40.20)$$

such that a vector $\vec{x}$ is rotated into $\tilde{x} = \vec{x} + \vec{\xi}(\vec{x})$, with $\vec{\xi}(\vec{x}) = \epsilon(-y, x, 0)^T$, and with $x^0 \rightarrow \tilde{x}^0 = x^0$, i.e. x^0 does not transform. The same conclusion is reached if one considers rotations around the yz and zx axes, and a general rotation is a composition of these three rotations. From these considerations we conclude that, under (infinitesimal) spatial rotations,

$$h_{00}(x) \rightarrow \tilde{h}_{00}(x) = h_{00}(x) - 2\partial_0 \xi_0(x) = h_{00}(x), \quad (40.21)$$

i.e. $h_{00}(x)$ does not transform under spatial rotations.

Next, under spatial rotations h_{0i} transforms as,

$$h_{0i}(x) \rightarrow \tilde{h}_{0i}(\tilde{x}) = R_i^j h_{0j}(x) \quad (40.22)$$

which is a vectorial transformation and h_{ij} transforms as,

$$h_{ij}(x) \rightarrow \tilde{h}_{ij}(\tilde{x}) = R_i^k R_j^l h_{kl}(x) \quad (40.23)$$

which is how a direct product of two vectors, $T = V_1 \otimes V_2$, transforms, and T is a $(2, 0)$ tensor under spatial rotations. The infinitesimal form of the above transformations is easily inferred from (40.20).

(b) h_{0i} contains 3 degrees of freedom, S one and B_i two, therefore the decomposition (40.3) preserves the number of degrees of freedom. Similarly, in Eq. (40.4), h_{ij} contains 6 degrees of freedom, each of the two scalars ψ and E one, F_j two and h_{ij}^{TT} two, such that on both sides of equation (40.4) there are six degrees of freedom.

(c) From Eq. (40.7) we see that

$$\delta h_{00} = \tilde{h}_{00} - h_{00} = -2\partial_0 \xi_0 \implies \delta \phi = -\partial_0 \xi_0, \quad (40.24)$$

which agrees with (40.8). Next,

$$\begin{aligned} \delta h_{0i} &= \tilde{h}_{0i} - h_{0i} = -\partial_0 \xi_i - \partial_i \xi_0 \implies \delta [B_i + \partial_i S] = -\partial_0 [\xi_i^T + \partial_i \xi^S] - \partial_i \xi_0 \\ \implies \delta B_i &= -\partial_0 \xi_i^T, \quad \delta S = -\partial_0 \xi^S - \xi_0, \end{aligned} \quad (40.25)$$

which agrees with Eqs. (40.12) and (40.9). Finally,

$$\begin{aligned} \delta h_{ij} &= \tilde{h}_{ij} - h_{ij} = -\partial_i \xi_j - \partial_j \xi_i \\ \implies \delta [2\delta_{ij}\psi + \partial_i \partial_j E + \partial_i F_j + \partial_j F_i + h_{ij}^{TT}] &= -\partial_i [\xi_j^T + \partial_j \xi^S] - \partial_j [\xi_i^T + \partial_i \xi^S] \\ \implies \delta \psi &= 0, \quad \delta E = -\xi^S, \quad \delta F_i = -\xi_i^T, \quad \delta h_{ij}^{TT} = 0, \end{aligned} \quad (40.26)$$

which agrees with Eqs. (40.10), (40.11), (40.13) and (40.14).

(d) From

$$\delta[\phi] = -\partial_0 \xi_0, \quad \delta[\partial_0 S] = -\partial_0 \xi_0 - \partial_0^2 \xi^S, \quad \delta[\partial_0^2 E] = -\partial_0^2 \xi^S \implies \delta[\phi - \partial_0 S + \partial_0^2 E] = 0, \quad (40.27)$$

we conclude that the Bardeen potential, $\Phi = \phi - \partial_0 S + \partial_0^2 E$ is invariant under linear coordinate transformations, *i.e.* Φ is gauge invariant potential. The two gauge invariant potentials, $\Psi = \psi$ and Φ are known as Bardeen potential, after James M. Bardeen who in 1981 first constructed them, James Bardeen is a son of the Nobel Prize laureate John Bardeen, one of the authors of the BCS theory of superconductivity.

Analogously one can construct a gauge invariant vector from,

$$\delta[B_i] = -\partial_0 \xi_i^T, \quad \delta[\partial_0 F_i] = -\partial_0 \xi_i^T \implies \delta[B_i - \partial_0 F_i] = 0, \quad (40.28)$$

from which we conclude that, $\bar{B}_i = B_i - \partial_0 F_i$ is a gauge invariant transverse vector. In conclusion, we have constructed two gauge invariant scalars (each of which counts as one degree of freedom), one gauge invariant vector (which has two degrees of freedom) and one gauge

invariant tensor (which has two degrees of freedom). In total this makes six gauge invariant degrees of freedom. We have started with 10 degrees of freedom of $h_{\mu\nu}$ and used four components of the gauge transformations ξ^μ to construct gauge invariant quantities which contain in total six (gauge invariant) degrees of freedom. Recall that Einstein's gravity is a nonlinear theory, and all that we have done in this problem applies only to the theory of linear perturbations. Because of the complexity in construction of gauge invariant quantities at higher than linear order in perturbations, the use of gauge invariant quantities is limited to the linear and quadratic theory of perturbations. Finally, above we have constructed the two gauge invariant Bardeen potentials Φ and Ψ . Any linear combination of these two potentials, $\alpha\Phi + \beta\Psi$ (with α and β real numbers) is also a gauge invariant scalar quantity.

(e) Upon acting with $\eta^{\mu\rho}$ on $R_{\mu\nu\rho\sigma}$ in Eq. (40.15) one obtains $R_{\nu\sigma}$ in Eq. (40.16).

(f) Eq. (40.16) can be decomposed into the (00) , $(0i)$ and (ij) components (due to the symmetry of $R_{\mu\nu}$ the $(i0)$ component is not independent). Let us first consider the R_{00} ,

$$\begin{aligned} R_{00} &= \partial_0 \partial^\alpha h_{\alpha 0} - \frac{1}{2} \partial^2 h_{00} - \frac{1}{2} \partial_0^2 h = -\partial_0^2 h_{00} + \partial_0 \partial_i h_{i0} - \frac{1}{2} (-\partial_0^2 + \partial_j^2) h_{00} - \frac{1}{2} \partial_0^2 (-h_{00} + h_{ii}) \\ &= -\frac{1}{2} \nabla^2 h_{00} + \partial_0 \partial_i h_{i0} - \frac{1}{2} \partial_0^2 (h_{ii}) = -\nabla^2 \phi + \partial_0 \nabla^2 S - 3\partial_0^2 \psi - \partial_0^2 \nabla^2 E \\ &= -\nabla^2 [\phi - \partial_0 S + \partial_0^2 E] - 3\partial_0^2 \psi = -\nabla^2 \Phi - 3\partial_0^2 \Psi, \end{aligned} \quad (40.29)$$

where here $\nabla^2 = \partial_j^2$ denotes the flat space Laplacian.

Analogously, we have for R_{0i} ,

$$\begin{aligned} R_{0i} &= \frac{1}{2} \partial_0 \partial^\alpha h_{\alpha i} + \frac{1}{2} \partial_i \partial^\alpha h_{\alpha 0} - \frac{1}{2} \partial^2 h_{0i} - \frac{1}{2} \partial_0 \partial_i h \\ &= -\frac{1}{2} \nabla^2 h_{0i} + \frac{1}{2} \partial_i \partial_j h_{j0} - \frac{1}{2} \partial_0 \partial_i (h_{jj}) + \frac{1}{2} \partial_0 \partial_j (h_{ji}) \\ &= -\frac{1}{2} \nabla^2 [B_i + \partial_i S] + \frac{1}{2} \partial_i \partial_j [B_j + \partial_j S] - \frac{1}{2} \partial_0 \partial_i [6\psi + 2\nabla^2 E] \\ &\quad + \frac{1}{2} \partial_0 \partial_j [2\delta_{ij} \psi + 2\partial_i \partial_j E + \partial_i F_j + \partial_j F_i + h_{ij}^{TT}] \\ &= -\frac{1}{2} \nabla^2 [B_i - \partial_0 F_i] - 2\partial_0 \partial_i \psi = -\frac{1}{2} \nabla^2 \bar{B}_i - 2\partial_0 \partial_i \Psi. \end{aligned} \quad (40.30)$$

The last set of components are R_{ij} ,

$$\begin{aligned} R_{ij} &= -\frac{1}{2} \partial_i \partial_0 h_{0j} + \frac{1}{2} \partial_i \partial_l h_{lj} - \frac{1}{2} \partial_j \partial_0 h_{0i} + \frac{1}{2} \partial_j \partial_l h_{li} + \frac{1}{2} (\partial_0^2 - \nabla^2) h_{ij} - \frac{1}{2} \partial_i \partial_j (-h_{00} + h_{ll}) \\ &= \frac{1}{2} \partial_i \partial_j [2\phi] - \frac{1}{2} \partial_0 \partial_i [B_j + \partial_j S] - \frac{1}{2} \partial_0 \partial_j [B_i + \partial_i S] \\ &\quad + \frac{1}{2} \partial_i \partial_l [2\delta_{lj} \psi + 2\partial_l \partial_j E + \partial_l F_j + \partial_j F_l + h_{lj}^{TT}] + \frac{1}{2} \partial_j \partial_l [2\delta_{li} \psi + 2\partial_l \partial_i E + \partial_l F_i + \partial_i F_l + h_{li}^{TT}] \\ &\quad + \frac{1}{2} (\partial_0^2 - \nabla^2) [2\delta_{ij} \psi + 2\partial_i \partial_j E + \partial_i F_j + \partial_j F_i + h_{ij}^{TT}] - \frac{1}{2} \partial_i \partial_j [6\psi + 2\nabla^2 E] \\ &= \partial_i \partial_j [\phi - \partial_0 S + \partial_0^2 E] - \frac{1}{2} \partial_0 [\partial_i B_j - \partial_0 \partial_i F_j + \partial_j B_i - \partial_0 \partial_j F_i] - \partial_i \partial_j \psi - \delta_{ij} \partial^2 \psi - \frac{1}{2} \partial^2 h_{ij}^{TT} \\ &= \partial_i \partial_j [\Phi - \Psi] - \partial_0 \partial_{(i} \bar{B}_{j)} - \delta_{ij} \partial^2 \Psi - \frac{1}{2} \partial^2 h_{ij}^{TT}. \end{aligned} \quad (40.31)$$

Since $R_{\mu\nu}$ is gauge invariant, so is $R = \eta^{\mu\nu} R_{\mu\nu}$ and therefore also, $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} R$ is gauge invariant. Therefore, if the components of $T_{\mu\nu}$ are gauge invariant under linear coordinate transformations, then the Einstein equations, $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$, is also gauge invariant.

This property of the Einstein equation is special for linear coordinate transformations around Minkowski background, and does not hold on general backgrounds and at higher orders in coordinate transformations.

(g) The Ricci scalar equals,

$$R = \eta^{\mu\nu} R_{\mu\nu} = -R_{00} + \delta_{ij} R_{ij} = 2\nabla^2 \Phi + 2(3\partial_0^2 - 2\nabla^2) \Psi. \quad (40.32)$$

The component of the Einstein tensor are then,

$$G_{00} = R_{00} + \frac{1}{2}R = -2\nabla^2 \Psi \quad (40.33)$$

$$G_{0i} = R_{0i} = -\frac{1}{2}\nabla^2 \bar{B}_i - 2\partial_0 \partial_i \Psi \quad (40.34)$$

$$G_{ij} = R_{ij} + \frac{1}{2}\delta_{ij}R = \partial_i \partial_j [\Phi - \Psi] - \partial_0 \partial_{(i} \bar{B}_{j)} - \frac{1}{2}\partial^2 h_{ij}^{TT} + \delta_{ij} \nabla^2 \Phi + \delta_{ij} (4\partial_0^2 - 3\nabla^2) \Psi, \quad (40.35)$$

such that all components of $G_{\mu\nu}$ are clearly gauge invariant. From Eq. (40.33) we see that – in the vacuum,

$$\nabla^2 \Psi = \nabla^2 \psi = 0, \quad (40.36)$$

which is in accordance with Eq. (40.17). That means that, by setting the right boundary conditions at the spatial infinity, one gets $\Psi = \text{const.}$ everywhere. Next, from Eq. (40.34) we see that,

$$\nabla^2 \bar{B}_i = 0, \quad (40.37)$$

such that – by the right choice of boundary conditions at the spatial infinity ($\bar{B}_i(\|\vec{r}\| \rightarrow \infty) \rightarrow 0$), the (gauge invariant) vector perturbation can be set to zero everywhere. Finally, let us consider the (ij) equation (40.35). First note that this equation can be split into two independent equations (by making use of the longitudinal and transverse projectors, which are defined by, $P_{ij}^L = \partial_i \partial_j / \nabla^2$ and $P_{ij}^T = \delta_{ij} - \partial_i \partial_j / \nabla^2$. These projectors are idempotent and mutually orthogonal, $P^L \cdot P^L = P^L$, $P^R \cdot P^R = P^R$, $P^L \cdot P^R = 0 = P^R \cdot P^L$). Thus equation (40.35) contains two independent pieces of information,

$$\partial_i \partial_j (\Phi - \Psi) = 0, \quad \partial^2 h_{ij}^{TT} = 0 \quad (40.38)$$

While the first equation tells us that, $\Phi = \Psi = 0$, the second is the wave equation for gravitational waves in the traceless, transverse gauge, in which the gravitational wave amplitude h_{ij}^{TT} is gauge invariant (under infinitesimal coordinate shifts). When one considers gravitational waves beyond the linear theory, h_{ij}^{TT} is not any more gauge invariant, and this gauge choice is known as harmonic gauge, and it is defined by, $h_{0\mu} = 0$, $\partial_i h_{ij} = 0$, $\delta_{ij} h_{ij} = 0$ (there is also the curved background version of these conditions). Many aspects of this problem are discussed in appendix B of reference <https://arxiv.org/pdf/1101.5059.pdf>, even though the decomposition used there is somewhat different.